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CO4-AT1 CODING CHALLENGE

A system must construct a network that connects all nodes with the least total cost, starting from a given node.

Using Prim's algorithm, iteratively select the minimum weight edge to expand the tree until all nodes are connected.

Sample Input:

n = 4

Graph =

0 1 4 0

1 0 2 5

4 2 0 3

0 5 3 0

Expected Output:

MST Cost = 6

ANSWER;

CODE;

```
#include <stdio.h>
```

```
#define INF 999
int main() {
 int n = 4;
  int graph[4][4] = {
    {0, 1, 4, 0},
    {1, 0, 2, 5},
    {4, 2, 0, 3},
    {0, 5, 3, 0}
  };
  int visited[4] = {0};
  int edges = 0, cost = 0;
  visited[0] = 1;   // Start from node 0
  while (edges < n - 1) {
    int min = INF;
    int x = -1, y = -1;
    for (int i = 0; i < n; i++) {
      if (visited[i]) {
        for (int j = 0; j < n; j++) {
          if (!visited[j] && graph[i][j] != 0 &&
```

```
                graph[i][j] < min) {
                min = graph[i][j];
                x = i;
                y = j;
              }
            }
          }
        }
        visited[y] = 1;
        cost += min;
        edges++;
        printf("Edge: %d - %d  Cost: %d\n", x, y, min);
    }
    printf("MST Cost = %d\n", cost);
    return 0;
}
```

Output

Edge: 0 - 1 Cost: 1

Edge: 1 - 2 Cost: 2

Edge: 2 - 3 Cost: 3

MST Cost = 6

Logic

- Start at node **0**
- Select minimum edge: **0–1 = 1**
- Select minimum edge: **1–2 = 2**
- Select minimum edge: **2–3 = 3**
- Total = **1 + 2 + 3 = 6**

Therefore, MST Cost = 6.